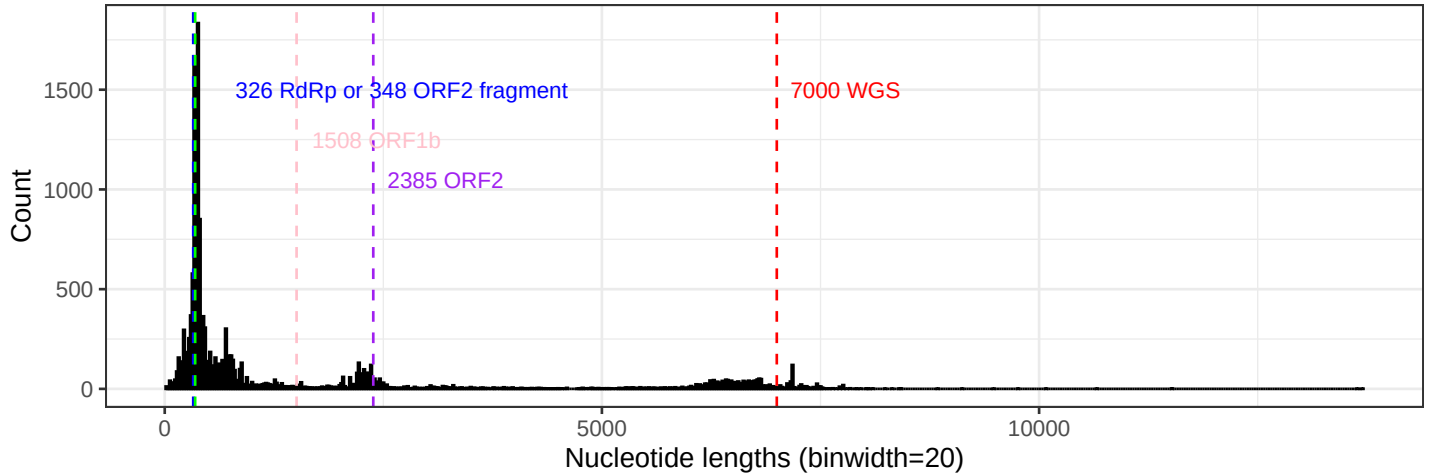


Exploratory Graphics

What is the genome length of astrovirus?

The astrovirus genome is approximately 6 to 8kb in length.

Diagnostic plot to estimate min-length filter for phylogenetic analysis (all data = 16381)

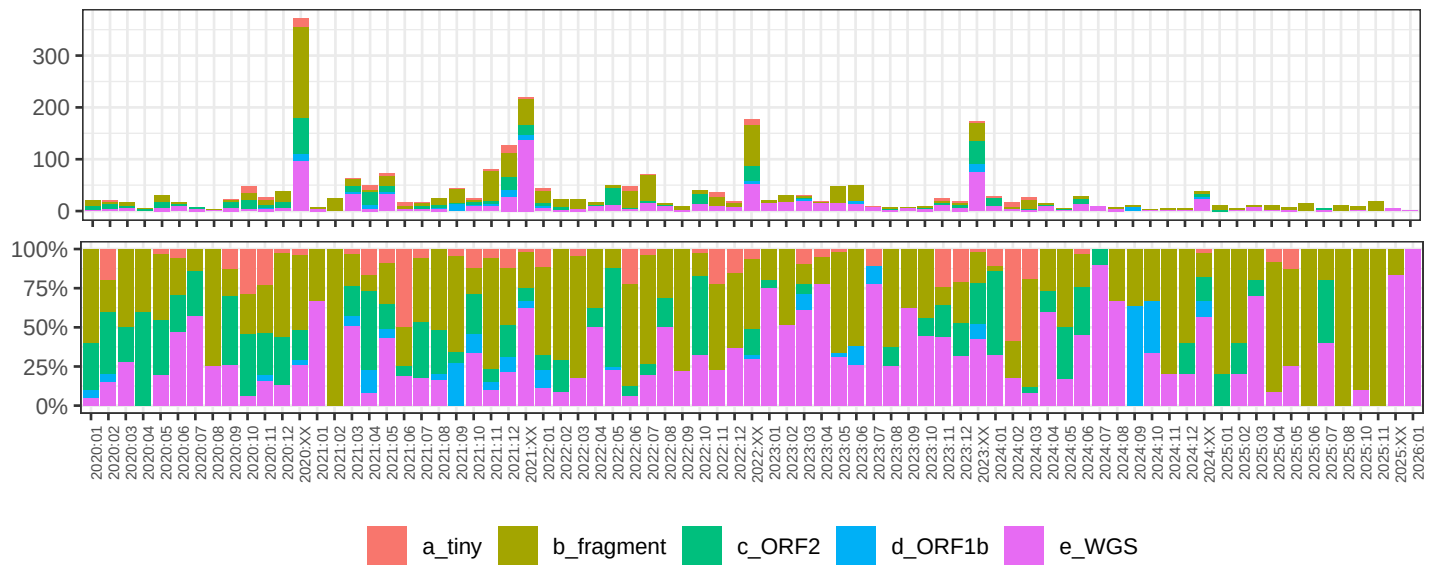


What is the distribution by length and host type?

```
cdata <- data %>%
  subset(., !is.na(date)) %>%
  subset(., date != "XXXX-XX-XX") %>%
  mutate(
    date_adjusted = lubridate::date(gsub("-XX", "-01", date)),
    col_year = substr(date, 1, 4),
    col_mon = substr(date, 6, 7),
    col_day = substr(date, 9, 10),
    len_category = case_when(length > 4000 ~ "e_WGS",
                             length > 1700 ~ "c_ORF2",
                             length > 1000 ~ "d_ORF1b",
                             length > 300 ~ "b_fragment",
                             TRUE ~ "a_tiny"),
    host_category = case_when(host %in% c("Homo sapiens") ~ "Z_Human",
                              is.na(host) ~ "NA",
                              TRUE ~ "A_Other"),
    host_length = paste0(host_category, "_", len_category)
  )

diagnostic_time_fill_plot_month(
  subset(cdata, col_year > 2019),
  margin_l=0.5, fill="len_category"
)
```

Frequency and proportion (n=2731)

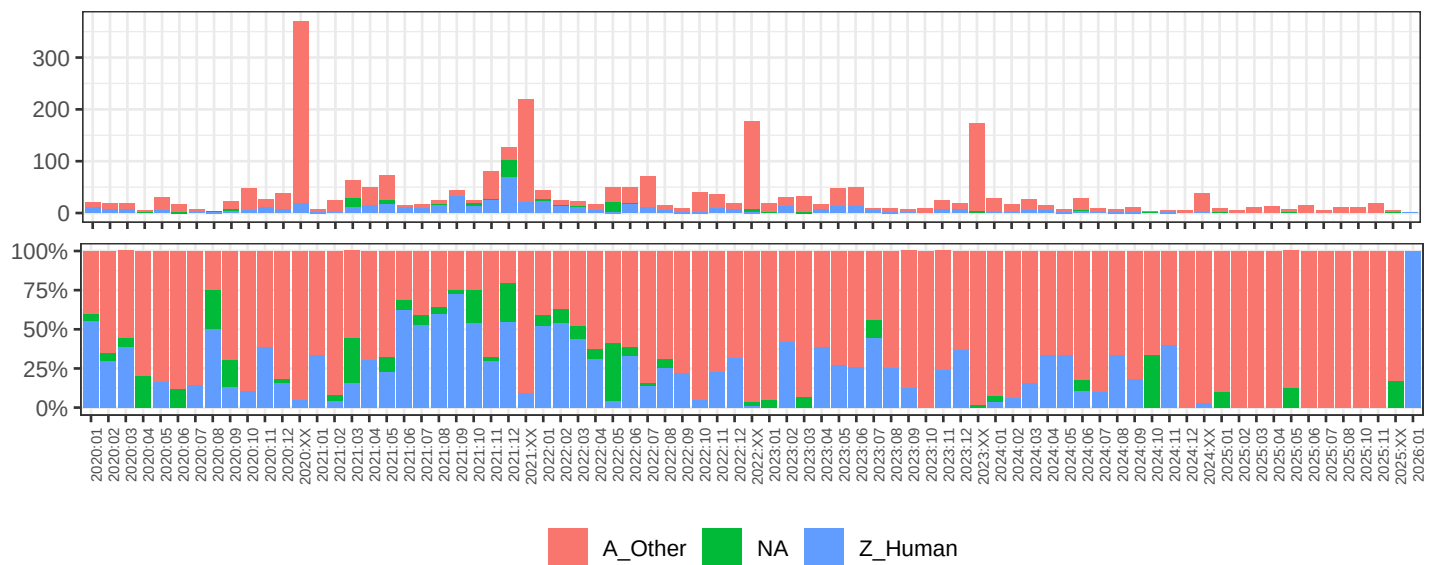


```
cdata %>% filter(col_year > 2019) %>% group_by(len_category) %>% summarise(n = n(), .groups = "drop")
```

```
## # A tibble: 5 x 2
##   len_category    n
##   <chr>        <int>
## 1 a_tiny      185
## 2 b_fragment  1103
## 3 c_ORF2      484
## 4 d_ORF1b     123
## 5 e_WGS      836
```

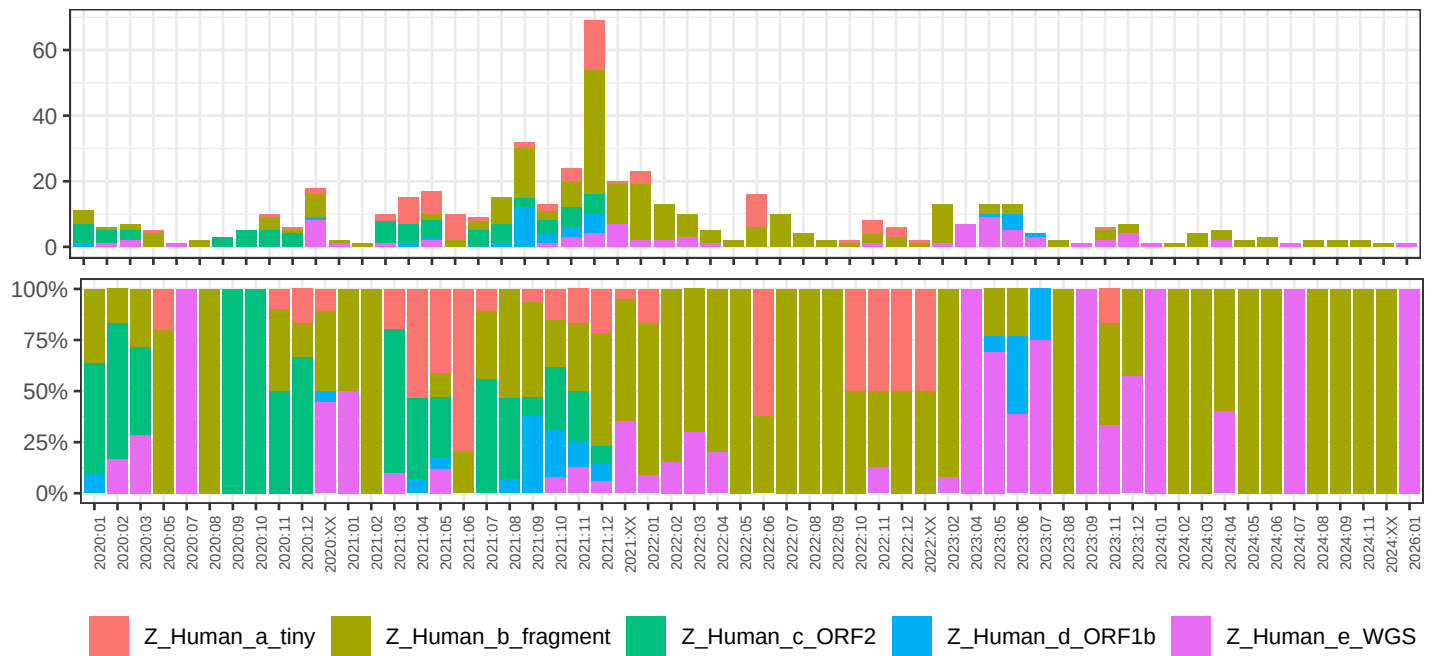
```
diagnostic_time_fill_plot_month(
  subset(cdata, col_year > 2019),
  margin_l=0.5, fill="host_category"
)
```

Frequency and proportion (n=2731)



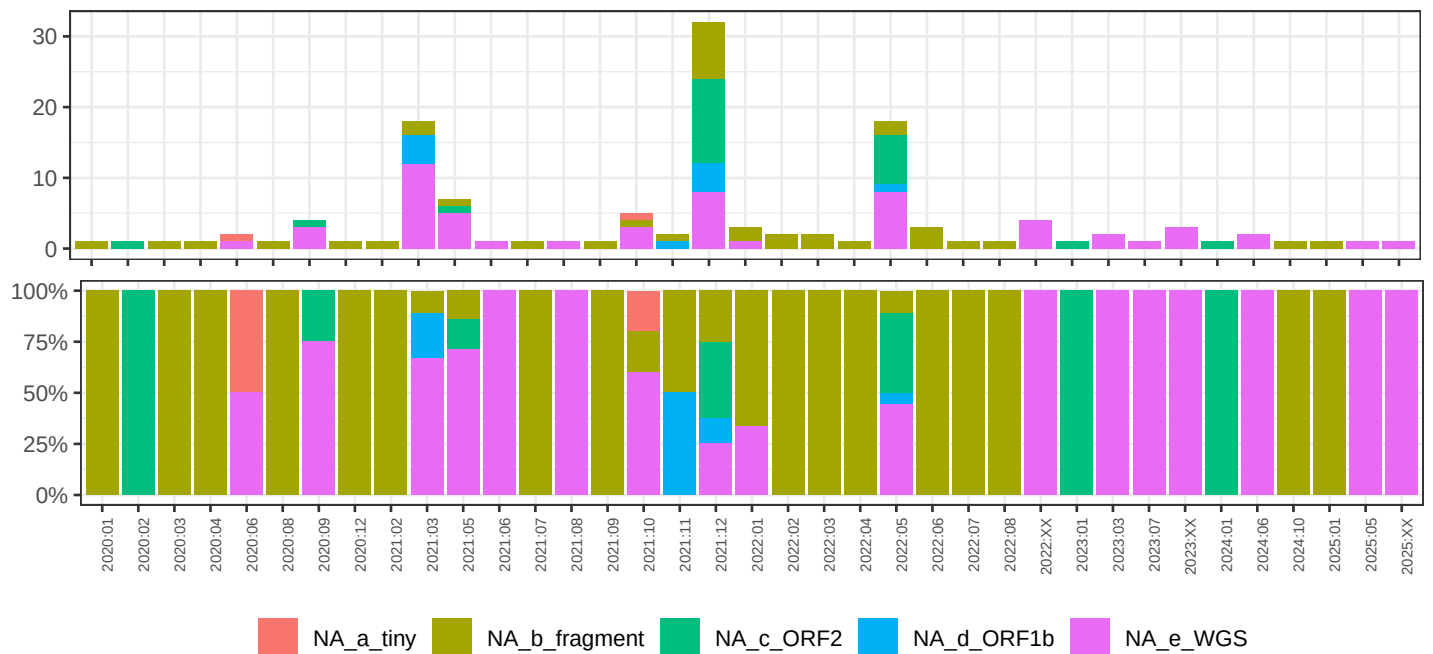
Focused on human samples:

Frequency and proportion (n=505)

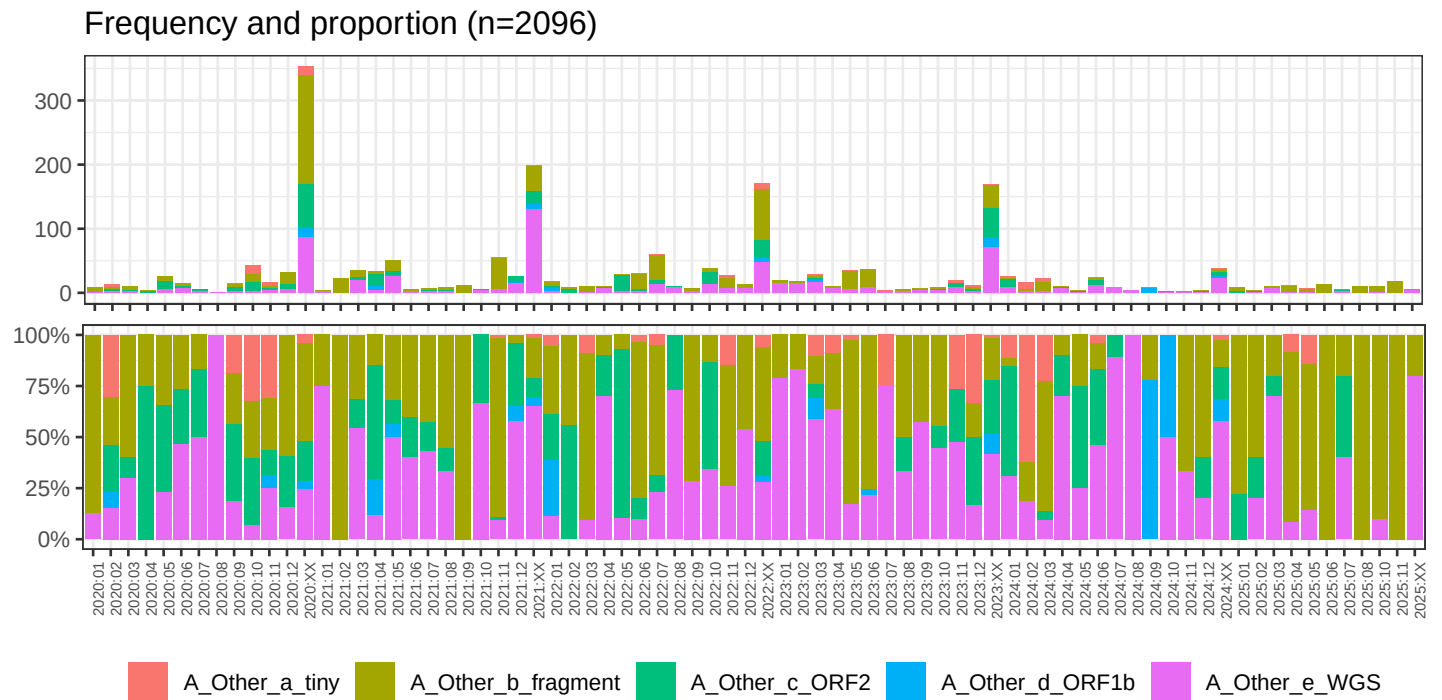


Focused on "NA" samples:

Frequency and proportion (n=130)



Focused on "Other" samples:



Who are the main submitters of sequence data?

Table 1: Top 25 most frequent sequence submitters with their region and countries

authors	n	regions	countries
?	750	Asia, North America, NA, Africa, South America	China, USA, NA, Malaysia, Bangladesh, Republic
tao et al.	709	Asia	China
armero et al.	425	Asia	China, Vietnam
li et al.	340	Asia, North America	China, USA, Canada
chen et al.	323	Asia	China
wang et al.	311	Asia, Africa, North America, NA	China, Kenya, Uganda, USA, NA
qiu et al.	302	Asia	China
wu et al.	259	Asia	China, Thailand, Taiwan
zhou et al.	257	Asia, Europe	China, Spain
he et al.	254	NA, Asia	NA, China, Myanmar
niendorf et al.	205	Europe	Germany, Hungary, Czech Republic
chu et al.	199	Asia	Hong Kong, Vietnam, Sri Lanka, Cambodia
yoshitomi et al.	187	Asia	Japan
pativada et al.	181	Asia	India
lee et al.	174	Asia, NA	South Korea, NA
zhirakovskaia et al.	166	Europe	Russia
yan et al.	165	Asia	China
zhang et al.	162	Asia, North America, Africa	China, USA, Sierra Leone
ushijima et al.	160	Asia	Japan
jindal et al.	158	North America	USA
zhu et al.	157	Asia, NA	China, NA
chidoti et al.	155	Africa	Zimbabwe
sharp et al.	147	Europe	United Kingdom
guix et al.	145	Europe, NA, Asia	United Kingdom, NA, Spain, Albania, France, Net
wei et al.	144	Asia	Thailand, China

Table 2: Top 25 most frequent sequence submitters with their region and countries

authors	n	regions	countries
homo sapiens	4741	Europe, Asia, North America, Africa, South America, NA, Oceania	Italy, China, Thailand
NA	2222	Asia, Africa, Oceania, NA, North America, Europe, South America	China, Cameroon, N
sus scrofa	1462	Asia, South America, Europe, North America, Africa, Oceania, NA	China, Chile, Hunga
gallus gallus	765	Asia, Europe, Africa, South America, North America, NA, Oceania	China, Belgium, Jap
sus scrofa domesticus	560	Oceania, Africa, Europe, Asia, North America	Australia, South Afr
rattus norvegicus	537	Europe, Asia, Africa	Germany, Hungary,
bos taurus	500	Asia, Europe, North America, Oceania, Africa, South America	South Korea, Iran, C
meleagris gallopavo	400	NA, South America, Africa, North America, Europe	NA, Ecuador, Nigeri
anatidae	366	Asia, NA, Europe, North America	China, NA, Croatia,
canis lupus familiaris	347	Europe, South America, Asia, Oceania, North America	Greece, Hungary, Br
miniopterus schreibersii	207	Asia, Europe	China, Italy, South I
felis catus	182	Asia, North America, Oceania, Europe	Hong Kong, USA, A
miniopterus magnater	150	Asia	Hong Kong, Thailand
taphozous melanopogon	134	Asia	China, Thailand, Ca
mus musculus	130	North America, Asia, Africa, South America, Europe	USA, China, Madag
capra hircus	103	Asia, Europe, Africa, North America	China, Hungary, Ga
macaca mulatta	96	Asia, NA	Bangladesh, NA
rattus rattus	93	Africa, Asia	Madagascar, Réunio
aves	92	Asia, North America	China, India, USA
hipposideros	90	Africa, Asia	Zimbabwe, China, R
mormopterus francoismoutoui	84	Africa	Réunion
mus musculoides	84	Africa	Gabon
rattus tanezumi	84	Asia, Africa	China, Republic of t
columba livia	82	Asia, Africa, Europe	Kazakhstan, Réunio
eothenomys cachinus	75	Asia	China

What is the host/composition in each cluster?

major_cluster	cluster_size	hosts
AStV-SC-45	214	Anatidae, NA, Anser anser, Anser cygnoides, Anser sp., Gallus gallus
AStV-SC-3	177	Homo sapiens, NA, Zalophus californianus
AStV-SC-11	155	Bos taurus, Sus scrofa, Camelus dromedarius, Capra hircus, Budorcas tibetana, Camelus bactrianus, Capreolus capreolus,
AStV-SC-15	63	Sus scrofa, Bos taurus, Rattus norvegicus
AStV-SC-33	58	Canis lupus familiaris, NA, Canis lupus, Nyctereutes procyonoides
AStV-SC-21	35	Sus scrofa domesticus, Sus scrofa
AStV-SC-13	33	Gallus gallus
AStV-SC-14	33	Sus scrofa
AStV-SC-32	31	Bos taurus, Capra hircus, NA
AStV-SC-44	31	Anatidae, Anser sp., Anser, NA
AStV-SC-7	26	Cairina moschata, Anatidae, NA, Anas platyrhynchos
AStV-SC-1	21	Homo sapiens, NA
AStV-SC-30	19	Anatidae, Anas platyrhynchos, NA
AStV-SC-26	17	Bos taurus
AStV-SC-48	15	Gallus gallus, Neogale vison, Ptyas mucosa
AStV-SC-12	11	Columbidae, Spilopelia chinensis, Columba livia, NA
AStV-SC-9	11	Homo sapiens, NA
AStV-SC-20	10	Mus musculus, NA
AStV-SC-2	9	Homo sapiens, NA
AStV-SC-6	9	NA
AStV-SC-38	7	Rattus andamanensis, Rattus norvegicus
AStV-SC-22	6	Anatidae, Anas platyrhynchos
AStV-SC-27	6	Felis catus, Panthera tigris altaica
AStV-SC-40	6	Rattus norvegicus, Rattus andamanensis
AStV-SC-49	6	Mus musculus
AStV-SC-10	5	Sus scrofa
AStV-SC-113	5	Neogale vison, Meleagris gallopavo
AStV-SC-31	5	Felis catus, Canis lupus familiaris, NA
AStV-SC-16	4	Oryctolagus cuniculus
AStV-SC-19	4	Zalophus californianus
AStV-SC-28	4	NA
AStV-SC-43	4	Marmota himalayana
AStV-SC-46	4	Marmota himalayana
AStV-SC-99	4	Capra hircus
AStV-SC-103	3	Bubalus bubalis
AStV-SC-124	3	Capra hircus
AStV-SC-29	3	Homo sapiens
AStV-SC-4	3	NA, Mustela lutreola
AStV-SC-41	3	Bos taurus, NA
AStV-SC-52	3	Arnoglossus sp., Echelus uropterus, NA
AStV-SC-8	3	Homo sapiens, NA